

Oscar Deschamps

GitHub: github.com/JustThinkingHard

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Systems Security Engineer

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PROFILE

Master's student in Cybersecurity with 3 years of work-study experience in Linux and embedded environments. Specialized in low-level system software, kernel and driver development, eBPF/LSM, secure boot and reverse engineering, across both offensive and defensive security. Fluent in C, Assembly (x86/ARM) and Rust, from the bootloader up to userspace.

EXPERIENCE

- CentumTnS** Lyon
 - Embedded Systems Security Engineer (Apprenticeship)* December 2024 - September 2026
 - Kernel Development:** Analyzed and patched a critical Linux kernel crash (*dm-waitfor* boot parameter)
 - Driver Development:** Developed and patched Linux kernel drivers for an embedded platform
 - System Hardening:** Implemented Secure Boot and Chain of Trust (ARM TF-A, OP-TEE, U-Boot) with TPM provisioning on NXP i.MX 8M Plus
 - Trusted Execution:** Developed OP-TEE Trusted Applications and Linux/U-Boot drivers
 - OS Maintenance:** Built a Yocto BSP meta-layer; integrated A/B redundancy and atomic OTA updates
 - Lifecycle:** Performed CVE analysis and security maintenance across the product lifecycle
- LeNewt** Lyon
 - Embedded Systems Engineer* September 2023 - September 2024
 - System Programming:** Designed a C++/Qt HMI interfacing with system daemons (DBus, NetworkManager) for embedded network configuration
 - OTA Updates:** Architected and implemented an atomic update system (RAUC/SWUpdate) with A/B partitioning redundancy
 - Boot Security:** Hardened Yocto recipes and integrated boot-stage security measures

PERSONAL PROJECTS

- eBPF Ransomware Mitigation:** eBPF/LSM tool that detects and mitigates ransomware-class file writes in real time, without disrupting trusted processes
- Linux Kernel Rootkit (v6.x):** Advanced LKM: Kprobes/Ftrace hooking, CR0 WP bypass, direct shellcode injection via ptrace, and VFS manipulation for file hiding
- ELF Code-Cave Infector:** C tool that injects payloads into ELF code caves and relocates entry points for persistence
- Process Injector:** Runtime payload injection into a live process via ptrace
- HID Attack & CVE Exploitation:** Raspberry Pi Pico (Bad USB) chained to a Raspberry Pi, exploiting CVE-2025-6019 for unauthorized Linux root access
- Reverse Engineering:** Official Hackropole write-up published for the FCSC challenge “Sésame, ouvre-toi (harder)”
- Vulnerability Research:** Reverse-engineered an undocumented local IPC protocol in a commercial desktop client

SKILLS

- Kernel & OS Internals:** Linux kernel, eBPF/LSM, drivers, kernel patching, VFS, Device Tree, systemd, DBus, Yocto/Buildroot
- Systems Programming:** C, Rust, Assembly (x86/ARM), POSIX (fork, pipe, signals), ptrace, ELF binary parsing
- Security & Hardening:** Secure Boot, Chain of Trust, OP-TEE, Trusted Firmware-A (TF-A), TPM, RAUC, LSM
- Reverse Engineering & Debugging:** Static/dynamic analysis, ELF internals, x86/ARM disassembly, GDB, Ftrace
- Tooling:** Git, Make/CMake, C++, Qt, QThread management, CI/CD

EDUCATION

- ESGI** Lyon
 - Master in Cybersecurity – Offensive and Defensive security* Sept. 2024 - Sept. 2026
- Epitech** Lyon
 - Bachelor of Computer Science – C Programming, UNIX Systems, and Algorithmics* Sept. 2021 - Sept. 2024

MISCELLANEOUS

- Certification:** Cisco CCNA (Routing & Switching)
- Learning:** Reverse Engineering (x86/ARM), Rust, Malware knowledge
- Sport:** Basketball, Gym
- Other:** Video games, CTF & programming competitions